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Rhinometry and Rhinomanometry

Number: 0700

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This Clinical Policy Bulletin addresses rhinometry and rhinomanometry.

I. **Experimental, Investigational, or Unproven**

Aetna considers rhinomanometry, acoustic rhinometry, and optical rhinometry experimental, investigational, or unproven because of a lack of clinical studies demonstrating that these tests improve clinical outcomes and the effectiveness of these approaches has not been established.

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description

Information in the [brackets] below has been added for clarification purposes. Codes requiring a 7th character are represented by "+":

CPT codes not covered for indications listed in the CPB:

92512	Nasal function studies (e.g., rhinomanometry)

Other CPT codes related to the CPB:

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Code	Code Description
31231 - 31235	Nasal and nasal/sinus, diagnostic, endoscopy
70450 - 70470	Computed tomography, head or brain

ICD-10 codes not covered for indications listed in the CPB:

J01.00 - J01.91	Acute sinusitis
J30.1 - J30.9	Vasomotor and allergic rhinitis
J32.0 - J32.9	Chronic sinusitis
J34.2	Deviated nasal septum
J34.3	Hypertrophy of nasal turbinates

Background

Acoustic Rhinometry and Rhinomanometry

There are many potential causes for nasal obstruction. Some of the most common causes are allergic rhinitis, deviation of the nasal septum, or sinus or nasal infection. Nasal obstruction is typically diagnosed by a patient's subjective complaint of nasal stuffiness coupled with a physical examination demonstrating anatomic restriction of the nasal passages.

Rhinomanometry and acoustic rhinometry are objective tests that have been attempted to assess nasal airway patency. Rhinomanometry measures air pressure and the rate of airflow during breathing. These measurements are then used to calculate nasal airway resistance. Acoustic rhinometry uses a reflected sound signal to measure the cross-sectional area and volume of the nasal passage. Acoustic rhinometry gives an anatomic description of a nasal passage, whereas rhinomanometry gives a functional measure of the pressure/flow relationships during the respiratory cycle. Both techniques have been used in comparing decongestive action of antihistamines and corticosteroids and for assessment of an individual prior to or following nasal surgery.

Papon et al (1985) stated that nasal compliance is a measure related to the blood volume in the nasal mucosa. The objective of this study was to better understand the vascular response in vasomotor rhinitis by measuring nasal cross-sectional area (CSA) and nasal compliance before and after mucosal decongestion in 10 patients with vasomotor rhinitis compared with 10 healthy subjects. Nasal compliance was inferred by measuring nasal area by acoustic rhinometry (AR) at pressures ranging from atmospheric pressure to a negative pressure of -10 cm H2O. Mucosal decongestion was obtained with 1 puff per nostril of 0.05 % oxymetazoline. At atmospheric pressure, nasal CSA were similar in the vasomotor rhinitis group and the healthy subject group. Mucosal decongestion did not induce any decrease of nasal compliance in patients with vasomotor rhinitis in contrast with healthy subjects. The authors concluded that these findings supported the hypothesis, already proposed, of an autonomic dysfunction based on a paradoxical response of the nasal mucosa in vasomotor rhinitis. Moreover, the clearly different behavior between healthy subjects and vasomotor rhinitis subjects suggested that nasal compliance measurement may thus represent a potential line of research to develop a diagnostic tool for vasomotor rhinitis, which remains a diagnosis of exclusion.

Elbrond et al (1991) noted that AR is a new method that described the geometry of the nasal cavity and the epi-pharynx. The method, based on the reflection of an acoustic signal entered into the nasal cavity, can be used to evaluate the CSA of the nasal cavity as a function of distance from the nostril. The method has, together with nasal expiratory peak flow (NEPF) and nasal index based upon a self-assessment score, been used to evaluate, in an objective and dynamic way, the effect of systemic treatment of nasal polyps with steroids in a series of 8 patients with recurrent nasal polyposis. The study showed a significant relationship

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between these 3 parameters before and after systemic treatment of nasal polyps with steroids. The authors concluded that AR is an accurate and objective method for measuring the geometry of the nasal cavity before and after treatment for processes that block the nasal cavity. This was a small study (n = 8).

Some investigators have found that subjective symptoms as rated by patients frequently do not correlate with rhinomanometry and acoustic rhinometry measurements (Passali et al, 2000). In addition, significant symptoms can be present without airway restriction (e.g., in patients with atrophic mucosa or sinusitis). In a clinical trial (n = 49) on the role of acoustic rhinometry in the diagnosis of adenoidal hypertrophy, Riechelmann et al (1999) reported that acoustic rhinometry, in general, is not suitable for assessing adenoidal size in pre-school children. He found that acoustic rhinometry was not able to differentiate controls from symptomatic children admitted for adenoidectomy.

Pallanch et al (1998) stated that there are many objective test values where some patients will complain of obstruction but others will not. It follows that there is not a single population threshold for the airway at which symptomatic obstruction would occur. Instead, it appears that there is a range of individual threshold values. Thus, it is not always possible to identify who will feel obstructed based on airway data.

There is inadequate evidence of the clinical utility of rhinomanometry and acoustic rhinometry. These tests have not been demonstrated to be superior to physical examination, nasal endoscopy or computed tomography (CT) imaging in selecting patients who would benefit from medical and/or surgical management of their nasal obstruction. Clinical studies published in the peer-reviewed medical literature are necessary to determine the value of rhinomanometry and acoustic rhinometry in the diagnosis and clinical management of patients with nasal obstruction.

Tarhan et al (2005) compared acoustic rhinometry (AR) data to CT data to evaluate the accuracy of AR measurements in estimating nasal passage area and evaluated its ability of quantifying paranasal sinus volume and ostium size in live humans. Twenty nasal passages of 10 healthy adults were examined by using AR and CT. Actual cross-sectional areas of the nasal cavity, sinus ostia sizes, and maxillary and frontal sinus volumes were determined from CT sections perpendicular to the curved acoustic axis of the nasal passage. Nasal cavity volume (from nostril to choana) calculated from the AR-derived area-distance curve was compared with that from the CT-derived area-distance curve. AR measurements were also done on pipe models that featured a side branch (Helmholtz resonator of constant volume but 2 different neck diameters) simulating a paranasal sinus. In the anterior nasal cavity, there was good agreement between the cross-sectional areas determined by AR and CT. However, posterior to the sinus ostia, AR over-estimated cross-sectional area. The difference between AR nasal volume and CT nasal volume was much smaller than the combined volume of the maxillary and frontal sinuses. The results suggested that AR measurements of the healthy adult nasal cavity are reasonably accurate to the level of the paranasal sinus ostia. Beyond this point, AR over-estimates cross-sectional area and provides no quantitative data for sinus volume or ostium size. The effects of paranasal sinuses and acoustic resonances in the nasal cavity are not accounted for in the present AR algorithms.

Cakmak et al (2005) evaluated how anatomic variations of the nasal cavity affect the accuracy of AR measurements. A cast model of a human nasal cavity was used to examine the effects of the nasal valve and paranasal sinuses on AR measurements. A luminal impression of a cadaver nasal cavity was made, and a cast model was created from this impression. To simulate the nasal valve, inserts of varying inner diameter were placed in the model nasal passage. To simulate the paranasal sinuses, side branches with varying neck diameters and cavity volumes were attached to the model. The AR measurements of the anterior nasal passage were reasonably precise when the passage area of the insert was within the normal range. When the passage area of the insert was reduced, AR measurements significantly under-estimated the cross-sectional areas beyond the insert. The volume of the paranasal sinus had limited effect on AR measurements when the sinus ostium was small. However, when the ostium size was large, increasing the volume of the sinus led to significant over-estimation of AR-derived areas beyond the ostium. The authors concluded that the pathologies that narrow the anterior nasal passage result in the most significant AR error by causing area under-estimation beyond the constriction. It also appears that increased paranasal sinus volume causes over-estimation of areas posterior to the sinus ostium when the ostium size is large. If these physical effects are not considered, the results obtained during clinical examination with AR may be misinterpreted.

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Liu et al (2006) examined the association between AR findings and results of over-night polysomnography. Patients who were under the age of 20 years, had severe deviated nasal septum, had previously received nasal or palatal surgery, or could not complete sleep test or AR examination were excluded. Subjects' basic data including age, gender, neck circumference, and body mass index (BMI) were collected. All participants received AR before over-night polysomnography. The results along with sleep-test outcomes were recorded and analyzed. A total of 87 patients were included in this study. Patients with respiratory disturbance index (RDI) less than 5/hour (n = 26) or with RDI of 5 - 30/hour (n = 28) tended to have larger minimal cross-sectional area (MCA) compared with those of patients whose RDI was more than 30/hour (n = 33) (p = 0.001). A stepwise multiple regression analysis showed that BMI, male gender, and MCA were contributing factors in RDI. The R2 value of the multiple regression analysis was 0.406. The authors concluded that patients with severe obstructive sleep apnea tended to have smaller MCA when compared with patients with RDI less than 30/hour. However, it was hard to predict whether patients had obstructive sleep apnea from AR examination.

Bermuller and colleagues (2008) examined the diagnostic accuracy of rhinomanometry (RMM) and peak nasal inspiratory flow (PNIF) in functional rhino-surgery. Measurements were carried out on 40 healthy individuals and 53 patients with symptomatic nasal stenosis. Cut-offs for RMM and PNIF were defined by receiver operating characteristic analysis. A cut-off between normal and pathological of 700 ml/second for RMM at a trans-nasal pressure difference of 150 Pa, and of 2,000 ml/second (120 L/minute) for PNIF was calculated. No significant differences in terms of sensitivity of RMM and PNIF (0.77 versus 0.66), specificity (0.8 versus 0.8) and diagnostic accuracy (0.79 versus 0.72) were found. The authors concluded that RMM and PNIF provide valuable information to support clinical decision making. However, with both methods, about 25 % of symptomatic patients with functionally relevant nasal structural deformity were not detected. Furthermore, a negative test outcome of RMM or PNIF does not exclude a functionally relevant nasal stenosis.

Straszek and associates (2008) stated that despite a growing number of studies using AR in children, no reference material exists that incorporates the entire age and height interval of pre-school children up to puberty for a range of rhinometric variables. These researchers attempted to provide a reference range for nasal volumes and MCAs in healthy non-decongested children aged 4 to 13 years old. A total of 256 primary school children (mean age of 7.95 years; range of 3.8 to 13.1 years; 123 boys/133 girls) were measured by AR. Variables were MCA (first, second, and absolute minimum) and nasal volumes from 0 to 4 cm (Vol0-4), 0 to 5 cm (Vol0-5), 1 to 4 cm (Vol1-4), and 2 to 5 cm (Vol2-5) into the nasal cavity. Height and weight were measured and atopic status was determined by skin-prick test. Age as well as current and past respiratory health were recorded from a questionnaire. In multiple linear regression models, height was the main predictor for all AR variables although weight also was a significant predictor of MCAs. There was no association between any AR variables with sex, atopy, or hay fever; but children with current wheeze (within last 12 months) and asthma had decreased nasal patency. The authors concluded that this study presented the most extensive current reference material for AR in non-decongested pre-pubescent healthy children. They stated that the presented reference material will aid the interpretation and evaluation of future and present epidemiological studies based on AR in children.

Piszcz et al (2008) reported on the use AR in assessing nasal obstruction due to adenoid hypertrophy in patients referred for adenoidectomy; they also evaluated on changes in the volume of the nasopharynx after adenoidectomy. The examination was performed in patients (n = 30) aged 5 to 10 years with adenoid hypertrophy admitted for adenoidectomy. Ten children who are free of otolaryngological problems served as the control group. All subjects had AR performed and additionally, endoscopic method such as rhyno-fiberoscopy and endoscopy of nasopharynx were introduced in the patient's group. The study showed that children with adenoid hypertrophy have statistically significant reduction of nasopharyngeal volume (NPV) versus the control group. Adenoidectomy increases the NPV parameter and makes it equal to control group. The authors concluded that AR seems to be a promising method in the assessment of nasopharyngeal volume. They noted that this and further studies may help to reduce the number of "unnecessary" adenoidectomies, by making standards for NPV in different group of age.

In a clinical survey conducted between 2001 and 2007, Kjaergaard et al (2009) examined the relationship between nasal cavity dimensions and airflow based on measures of AR and peak nasal inspiratory flow

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(PNIF) in a very large sample of mixed rhinologic and non-rhinologic patients. The study population comprised 2,523 consecutive adult patients, mainly white, referred to the Department of Otolaryngology-Head and Neck Surgery, Sorlandet Hospital, Kristiansand, Norway, for evaluation of sleep-related disorders (e.g., snoring, sleep apnea) or chronic nasal complaints. Subjects underwent AR and PNIF at baseline and after decongestion of the nasal mucosa with xylometazoline hydrochloride. Questionnaires and height and weight measurements were obtained prior to the nasal recordings. Main outcome measure was associations between measures of AR (volume and area) and PNIF. Nearly linear relationships were found between PNIF in 4 categories and nasal cavity volumes and minimal CSA (analysis of variance, $p < 0.001$; post-hoc analysis, $p < 0.01$). Adjusted associations between 5 of 6 AR measures and PNIF both at baseline and after decongestion were significant ($p < 0.001$ in 9 cases and $p = 0.03$ in 1 case). The authors concluded that the findings of this study indicated statistically significant associations between nasal cavity dimensions and PNIF. The most important structural determinant for PNIF was the minimal CSA of the nasal cavity.

The authors stated that the study was based on a selected sample that was likely to differ from the general population in terms of nasal anatomy and physiological features. This was partly reflected by an over-representation of smokers and a preponderance of male sex due to a large number of male subjects with sleep apnea and snoring in the sample. Respiratory co-morbidity could potentially affect the PNIF measurements by limiting the inspiratory effort. Since asthma and allergy were self-reported, limitations were applicable to interpretation and extrapolation of the results. However, the prevalence of self-reported asthma and allergy in this study agreed well with reported prevalence. Another limitation was the lack of distinctions between asthma and chronic obstructive pulmonary disease (COPD).

Okun and colleagues (2010) evaluated the use of AR in children with obstructive sleep apnea (OSA). Subjects with clinically suspected OSA underwent AR measurements followed by attended over-night polysomnography. Of a total of 20 subjects (13 boys, 7 girls), 15 (75 %) had OSA, defined as apnea-hypopnea index (AHI) greater than or equal to 5 events per hour of sleep, and 5 had primary snoring (PS). The mean AHI was 16.79 versus 1.96 events/hour. Positional changes in airway measurement by AR were present in the OSA group, with an average decrease in nasal cavity volume from upright to supine position of 1.53 cm(3) ($p = 0.027$). These changes were predictive of sleep apnea ($r (2) = 0.65$, $p = 0.035$). The authors concluded that these findings showed a marked difference between OSA and PS groups during AR measurements of the nasopharynx. They stated that positional airway changes had been previously reported in adults with OSA and further evaluation of the airway function in pediatric OSA is warranted.

Andre and colleagues (2009) evaluated the correlation between the subjective sense of nasal patency and the outcomes found with rhinomanometry and AR. Review of English-language articles in which correlations were sought between subjective nasal patency symptoms and objective scores as found with rhinomanometry [nasal airway resistance (NAR)] and AR [minimal cross-sectional area (MCA)]. Correlations were related to unilateral or combined assessment of nasal passages and to symptomatic nasal obstruction or unobstructed nasal breathing. A total of 16 studies with a level of evidence II-a or II-b fit the inclusion criteria and were further analyzed. Almost every possible combination of correlations or lack thereof in relation to the variables included was found. However, when obstructive symptoms were present, a correlation between the patency symptoms with nasal airway resistance and minimal cross-sectional area was found more often than in the absence of symptoms. In cases of bilateral assessment a correlation was found almost as often as it was not between patency symptoms and total nasal airway resistance or combined minimal cross-sectional areas, while in the limited amount of studies in which unilateral assessment was done a correlation was found each time between patency symptoms and nasal airway resistance. The authors concluded that the correlation between the outcomes found with rhinomanometry and AR and an individual's subjective sensation of nasal patency remains uncertain. Based on this review, it seems that the chance of a correlation is greater when each nasal passage is assessed individually and when obstructive symptoms are present. There still seems to be only a limited argument for the use of rhinomanometry or AR in routine rhinologic practice or for quantifying surgical results.

Kupczyk et al (2010) evaluated AR as an objective method of assessment of nasal lysine aspirin (Lys-ASA) nasal challenge. A total of 20 patients with aspirin induced asthma (ASA-S) and 10 controls (ASA-NS): 5 patients with allergic rhinitis and 5 healthy subjects) were included. Nasal challenge was performed with placebo (saline) and 14.4 mg of Lys-ASA introduced as aerosol to both nostrils (total dose: 16 mg of

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acetylsalicylic acid). Measurements of nasal volume bilaterally were performed with the use of AR before and 1, 2, 4 and 24 hours after the challenge. For further analysis the sum of both nasal cavities volume at the level of 2 to 5 cm from nostrils was used. Mean total bilateral volume in ASA-S group after placebo was: 7.74, 6.21, 7.11, 7.12, 7.24 cm(3) and 7.24, 5.77, 6.31, 6.27, 6.98 cm(3) after Lys-ASA (before and after 1, 2, 4 and 24 hours, respectively; p = 0,048 and p = 0,02, in 2nd and 4th hour, Lys-ASA versus placebo, Wilcoxon's test). With cut-off point of nasal volume decrease by 10 % in the first hour the sensitivity of the test was 70 %, specificity 60 %, positive predictive value 77.78 % and negative predictive value 50 %. The authors concluded that AR with measurement of nasal cavities volume changes at 2 to 5 cm from nostrils does not appear to be sufficiently sensitive and specific as a single method for evaluation of studied challenge method.

In a prospective pilot study, Luong et al (2010) evaluated ORM as an objective evaluation of nasal patency using nasal provocation testing (NPT) with histamine and oxymetazoline. A total of 5 adult subjects with allergic rhinitis and 5 adult normal subjects who underwent challenge with histamine and oxymetazoline were included in this study. Patients underwent challenge with increasing concentrations of histamine to determine the amount of histamine needed to cause a positive ORM reading. The same subjects then underwent histamine challenge with this amount followed by oxymetazoline. Nasal patency was assessed subjectively after each challenge with the visual analog scale. The median histamine amount needed to cause a positive response was statistically lower in allergic rhinitis as compared with non-allergic subjects at 150 microg and 300 microg, respectively (p = 0.04). When comparing ORM with subjective nasal congestion after histamine and oxymetazoline challenges, there was a statistically significant correlation with r = 0.79 (p = 0.00003). The authors concluded that the findings of this pilot study demonstrated a correlation between subjective symptoms of nasal patency and objective measurements with ORM. Less histamine amount necessary to incite nasal congestion in allergic rhinitis suggests that these patients may be primed to the effects of histamine. They stated that these preliminary findings serve to create the foundation for further exploration of the utility of ORM for NPT.

Tombu et al (2010) stated that AR and RMM study 2 different parameters of nasal ventilation:

I. respiratory function and
II. the anatomy of nasal cavities.

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These researchers examined the usefulness of AR and RMM, in particular in the surgical field. They listed the normal values for these tests. Nasal obstruction is a symptom of multi-factorial origin. Nasal patency is only one factor influencing the sensation of nasal ventilation. Despite the range of divergent opinions in both the literature and among rhinological clinicians, the objective assessment of nasal patency in functional rhinoplasty or septo-rhinoplasty seems to be advisable. The authors stated that the roles of AR and RMM still have to be established.

de Aguiar Vidigal et al (2013) evaluated the nose of patients with OSA syndrome (OSAS), compared them to controls, and correlated the different methods used to evaluate the nose. A total of 47 patients with moderate-to-severe OSAS and 20 controls who were matched for gender, age, and BMI were included. Questionnaires regarding sleep and nasal symptoms, physical examination, AR, naso-fibroscopy, rhinoscopy, as well as nasal inspiratory peak flow (NIPF) measurements were performed. In the OSAS group, 33 (70.2 %) were male, with a mean age of 53.2 +/- 9.1 years. In the control group, 13 (65 %) were male, with a mean age of 53.7 +/- 9.7 years. The OSAS group had a higher score on the nasal symptoms scale (p < 0.01) and a higher frequency of nasal alterations [presence of septal deviation, clinical complaints (p = 0.01) and hypertrophy of the inferior nasal turbinate (p < 0.01)]. The NIPF and AR parameters could not differentiate between the OSAS and control groups. There were no significant correlations among the different methods used to evaluate the nose. Lower NIPF values were capable of predicting higher apnea-hypopnea index scores (p = 0.007). The authors concluded that clinical complaints and nasal alterations as measured by rhinoscopy and naso-fibroscopy were associated with the presence of OSAS, which was not the case for theAR and NIPF parameters. The results of different evaluation methods were not correlated with each other.

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Mendes et al (2012) correlated objective assessment of nasal obstruction, as measured by AR (volume of the first 5 cm of the nasal cavity) and active anterior RMM (total nasal airway resistance), with its subjective evaluation (obstruction scores). A total of 30 patients, aged 7 to 18 years, with persistent allergic rhinitis and 30 controls were enrolled. The obstruction score was reported for the whole nasal cavity and for each nostril separately. The 3 variables were measured at baseline and after induction of nasal obstruction. There were significant and negative correlations between resistance and nasal volume in all groups and scenarios, except for the most obstructed nostril, in the control group, post-obstruction. For the whole nasal cavity, there was no significant correlation between objective and subjective variables except between score and total nasal cavity volume in the control group, post-obstruction. Regarding the most obstructed nostril, these investigators found a significant negative correlation between score and resistance and a significant positive correlation between score and volume for the total group at baseline. There were no clear differences in the correlation coefficients found in patients and controls. The correlation coefficients did not change after induction of nasal obstruction. The authors concluded that objective assessment of nasal obstruction did not correlate significantly with subjective evaluation for the nasal cavity as a whole, but there was a correlation for unilateral assessments. There was correlation between the objective evaluations. Allergic rhinitis and acute induction of nasal obstruction did not affect the correlation between objective and subjective assessments of nasal obstruction. Moreover, they stated that addition of an objective method for evaluation of nasal obstruction could be useful in the research setting; if no such method can be used, each nostril should be evaluated separately.

Altuntas et al (2013) noted that Crimean-Congo hemorrhagic fever (CCHF), like other viral infections, may prolong muco-ciliary clearance time and increase nasal resistance in children. The aim of the present prospective case-control study was to study, using saccharin and anterior RMM tests, whether CCHF infections caused any change in nasal physiology. A total of 40 subjects (20 of whom had CCHF (group 1) and 20 of whom were healthy controls (group 2)), were enrolled in this study. The definitive diagnosis of CCHF infection was made based on typical clinical and epidemiological findings and detection of CCHF virus-specific IgM by ELISA or of genomic segments of the CCHF virus by reverse transcription-polymerase chain reaction. Anterior RMM was performed in all participants according to current recommendations of the Committee Report on Standardization of Rhinomanometry. A saccharin test was used to evaluate muco-ciliary clearance, and nasal muco-ciliary clearance time was assessed with the saccharin test as described previously. In these patients, the mean time from the application of saccharin crystals to the first feeling of a sweet taste was 6.77 ± 3.25 minutes (range of 2 to 16 mins). In terms of the mean time from the application of saccharin crystals to the first feeling of a sweet taste, there was no difference between 2 groups. The mean total air flow was 637.60 ± 76.18 ml/s (range of 490 to 760 ml/s). The mean total nasal airway resistance was 0.24 ± 0.03 Pa/ml/s (range of 0.20 to 0.31 Pa/ml/s). In terms of the degree of nasal air flow and nasal airway resistance and the total air flow and total nasal airway resistance of each nostril, there was no difference between the 2 groups. The authors concluded that the results obtained in anterior RMM and saccharin test showed that there was no statistically significant difference between CCHF (+) patients and controls. These findings suggested that CCHF virus infection does not affect nasal physiology. However, this is the first study performed on this issue and further studies on larger series need to be performed.

Haavisto and Sipila (2013) compared AR, RMM, and subjective estimation of the nasal obstruction before and after septoplasty and evaluated the long-term results of septal surgery. The study included 30 adult patients who were operated on because of septal deviation. Pre-operatively, AR and active anterior RMM were performed on each subject after decongestion of the nose. A visual analog scale (VAS) for unilateral nasal obstruction was filled in by the patients. The measurements were repeated both 6 months and 10 years post-operatively. A significant change in acoustic values was found during the long-term follow-up of 10 years. The mean minimal cross-sectional area on the more obstructive side was 0.35 cm(2) pre-operatively. Six months after operation, it was 0.52 cm(2), and 10 years after operation, it was 0.68 cm(2). The mean resistance fell from pre-operative 1.16 Pa/ml/s to 0.41 Pa/ml/s during the first 6 months, but rose again to 1.21 Pa/ml/s after 10 years. Despite a tendency of improvement, no statistically significant change was found between pre-operative and post-operative values in VAS. Six months after operation, 69 % of the patients were satisfied with the result, and after 10 years the amount of satisfied patients was 83 %. The authors found an increase in acoustic values, but an increase in nasal resistance in the long-term follow-up. Other factors than nasal area may have an impact on nasal resistance and the feeling of nasal obstruction. The small size on the sample interfered with the results.

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In a prospective study, Toros et al (2013) evaluated the differences in acoustic rhinometric findings between the affected and non-affected sides in patients with unilateral chronic otitis media (COM) and examined if unilateral COM correlates with the side of nasal obstruction. A total of 55 consecutive patients with unilateral COM were involved in this study. All patients were evaluated with AR, the Nasal Obstruction Symptom Evaluation (NOSE) scale, and measurement of their nasal muco-ciliary transport time. The mean cross-sectional area 1, mean cross-sectional area 2, volume 1, and volume 2 values were not different between the affected and non-affected sides ($p > 0.05$). The NOSE score had a reverse correlation with the mean cross-sectional area 2 ($p < 0.05$) and volume 2 ($p < 0.01$) of the affected side. Saccharin time was not correlated with the acoustic rhinometric values of the affected side ($p > 0.05$). The authors concluded that these findings did not support the hypothesis that unilateral COM is correlated with the side of nasal obstruction.

In a prospective study, Dadgarnia and colleagues (2013) used the objective parameters of AR and rhinomanometry to evaluate the effectiveness of septoplasty surgery. A total of 30 patients for septoplasty surgery were enrolled in this study; AR and rhinomanometry tests were performed on all patients both before and 3 months following the operation. he symptom recovery rate was recorded according to the patient's statements and anterior rhinoscopic examinations 3 months post-surgery. Data were analyzed using a t-test and chi-square tests in a SPSS package. A total of 26 of 30 patients returned for a post-surgery follow-up examination after 3 months. Patients were aged from 18 to 32 years (average of 25 years). In total, 69.2 % (18 patients) were satisfied with the results of the procedure. In addition, rhinomanometry resulted in a decrease in general nasal resistance if patients used decongestants ($p = 0.03$). However, the decrease was not significant before the use of decongestants ($p = 0.12$). Furthermore, according to the results from AR, there was an increase in the nasal cross-sectional area on both the narrow and wide sides after the operation ($p < 0.05$), although this increase was not so notable in the narrower side after using decongestants. There was, however, no significant relationship between the results from the objective tests and the patient's symptoms or clinical examinations ($p > 0.05$). The authors concluded that these findings showed that although the objective tests confirm an improvement in general nasal resistance and an increase in the nasal cross-sectional area after surgery, no unambiguous relationship between the patient's symptoms and the clinical examinations was observed. Therefore, such objective tests did not prove to be sufficient diagnostic criteria for the effectiveness of septoplasty.

Patuzzi and Cook (2014) described a simple and inexpensive method for monitoring nasal air flow resistance using measurement of the small-signal acoustic input impedance of the nasal passage, similar to the audiological measurement of ear drum compliance with acoustic tympanometry. The method requires generation of a fixed sinusoidal volume-velocity stimulus using ear-bud speakers, and an electret microphone to monitor the resultant pressure fluctuation in the nasal passage. Both are coupled to the nose via high impedance silastic tubing and a small plastic nose insert. The acoustic impedance is monitored in real-time using a laptop soundcard and custom-written software developed in LabView 7.0 (National Instruments). The compact, lightweight equipment and fast time resolution lends the technique to research into the small and rapid reflexive changes in nasal resistance caused by environmental and local neurological influences. The authors concluded that the acoustic impedance rhinometry technique has the potential to be developed for use in a clinical setting, where the need exists for a simple and inexpensive objective nasal resistance measurement technique.

Lange et al (2014) stated that chronic rhino-sinusitis (CRS) is a disease related to the nose and the para-nasal sinus as defined by the European Position Paper on Rhinosinusitis and Nasal Polyps (EPOS) criteria. The criteria include subjective symptoms, such as nasal obstruction, and objective findings by endoscopy. Acoustic rhinometry is an objective method to determine nasal cavity geometry. The technique is based on a sound pulse reflection analysis in the nasal cavity and determines cross-sectional areas as a function of distance as well as volume. Acoustic rhinometric measurements in persons recruited from the general population, with and without CRS based on the clinical EPOS criteria, were investigated. As part of a trans-European study, a total of 362 persons, comprising 91 persons with CRS and 271 persons without CRS, were examined by an otolaryngologist including rhinoscopy. Minimum cross-sectional area, distance to minimum cross-sectional area, and volume in the nasal cavity were measured by AR and all participants underwent PNIF and allergy test. A difference in AR was found before and after decongestion, but no difference was seen between CRS patients and controls. Positive correlation between AR and PNIF was found and AR was

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capable of identifying mucosal edema and septum deviation visualized by rhinoscopy. The authors concluded that AR, as a single instrument, was not capable of discriminating persons with CRS from persons without CRS in the general population.

Brockmann et al (2013) examined the diagnostic test accuracy (DTA) of different tests for OSA compared to polysomnography (PSG) in children. These investigators performed a systematic review according to DTA criteria published by the Cochrane Collaboration. Studies that compared any possible diagnostic test with PSG for diagnosing OSA were considered. Study quality assessment was conducted in each selected study and DTA measures recalculated by hand whenever possible. Excellent DTA was defined as positive likelihood ratio (PLR) greater than 10 and negative likelihood ratio (NLR) less than 0.1. These researchers identified 1,064 potentially relevant studies, of which 33 met inclusion criteria. Study quality was generally low; 5 studies fulfilled all quality criteria and 11 studies included more than 100 subjects. Included studies compared 40 different tests to PSG. Only 13 studies used the currently accepted definition for OSA (i.e., AHI greater than or equal to1). In these studies, PLR ranged from 1.017 to infinity, NLR from 0 to 1.089. Sleep lab-based PSG, urinary biomarkers, and rhinomanometry (1 study each) showed excellent DTA. The authors concluded that there is limited evidence concerning diagnostic alternatives to PSG for identifying OSA in children. However, PSG, urinary biomarkers, and rhinomanometry may be valid tests if their apparently high DTA is confirmed by subsequent studies.

Aziz et al (2014) performed a systematic review of measurement tools utilized for the diagnosis of nasal septal deviation (NSD). Electronic database searches were performed using MEDLINE (from 1966 to second week of August 2013), EMBASE (from 1966 to 2nd week of August 2013), Web of Science (from 1945 to 2nd week of August 2013) and all Evidence Based Medicine Reviews Files (EBMR); Cochrane Database of Systematic Review (CDSR), Cochrane Central Register of Controlled Trials (CCTR), Cochrane Methodology Register (CMR), Database of Abstracts of Reviews of Effects (DARE), American College of Physicians Journal Club (ACP Journal Club), Health Technology Assessments (HTA), NHS Economic Evaluation Database (NHSEED) till the 2nd quarter of 2013. The search terms used in database searches were 'nasal septum', 'deviation', 'diagnosis', 'nose deformities' and 'nose malformation'. The studies were reviewed using the updated Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) tool. Online searches resulted in 23 abstracts after removal of duplicates that resulted from overlap of studies between the electronic databases. An additional 15 abstracts were excluded due to lack of relevance. A total of 8 studies were systematically reviewed. The authors concluded that diagnostic modalities such as acoustic rhinometry, rhinomanometry and nasal spectral sound analysis may be useful in identifying NSD in anterior region of the nasal cavity, but these tests in isolation are of limited utility. They stated that compared to anterior rhinoscopy, nasal endoscopy, and imaging the above mentioned index tests lack sensitivity and specificity in identifying the presence, location, and severity of NSD.

Yuksel (2014) investigated the effects of anterior rhinomanometry-induced nasal resistance on OSAS patients. Between May 2011 and September 2011, a total of 100 patients (76 males, 24 females; mean age of 47.6 ± 11.6 years; range of 20 to 71 years) who were admitted with complaints of snore, breathing pauses told by their partners, oversleep mood in a daytime and fatigue and diagnosed with OSAS by PSG with simple snore were included. Anterior rhinomanometry was applied for all patients and nasal resistance was estimated. Mallampati index and BMI of patients was calculated. The mean AHI and minimum oxygen saturation values were measured. There was no significant relationship between nasal resistance and AHI. However, a significant relationship between AHI and Mallampati and BMI values was observed. The AHI values increased, as the Mallampati and BMI values increased. The authors concluded that these findings showed that nasal resistance has no significant effect on AHI and minimum oxygen saturation in OSAS patients.

Major et al (2014) conducted a systematic review to examine the accuracy of alternative tests compared with naso-endoscopy (reference standard) for screening adenoid hypertrophy. The review included searches of electronic databases, hand-searches of bibliographies of relevant articles and gray literature searches. They included all articles in which an alternative test was compared with naso-endoscopy in children with suspected nasal or nasopharyngeal airway obstruction. These researchers identified 7 articles that were of poor to good quality. They identified the following alternative tests: multi-row detector computed tomography (sensitivity, 92 %; specificity, 97 %), videofluoroscopy (sensitivity, 100 %; specificity, 90 %),

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rhinomanometry with decongestant (sensitivity, 83 %; specificity, 83 %) and clinical examination (sensitivity, 22 %; specificity, 88 %). Lateral cephalograms tended to have good to fair sensitivity (typically 61 to 75 % and poor specificity (41 to 55 %) when adenoid size was evaluated but excellent to good specificity when airway patency was evaluated (68 to 96 %). The authors concluded that no ideal tool exists for dentists to screen adenoid hypertrophy, owing to access constraints, radiation concerns and suboptimal diagnostic accuracy. They stated that research is needed to identify a low-risk, easily acceptable, highly valid diagnostic screening tool.

Melo et al (2015) noted that when there is a change in the physiological pattern of nasal breathing, mouth breathing may already be present. The diagnosis of mouth breathing is related to nasal patency. One way to access nasal patency is by AR. These investigators systematically reviewed the effectiveness of AR for the diagnosis of patients with mouth breathing. Electronic databases LILACS, MEDLINE via PubMed and Bireme, SciELO, Web of Science, Scopus, PsycInfo, CINAHL, and Science Direct, from August to December 2013, were consulted. A total of 11,439 articles were found: 30 from LILACS, 54 from MEDLINE via Bireme, 5,558 from MEDLINE via PubMed, 11 from SciELO, 2,056 from Web of Science, 1,734 from Scopus, 13 from PsycInfo, 1,108 from CINAHL, and 875 from Science Direct. Of these, 2 articles were selected. The heterogeneity in the use of equipment and materials for the assessment of respiratory mode in these studies revealed that there is not yet consensus in the assessment and diagnosis of patients with mouth breathing. The authors concluded that according to the articles, AR has been used for almost 20 years, but controlled studies attesting to the effectiveness of measuring the geometry of nasal cavities for complementary diagnosis of respiratory mode are warranted.

In a prospective clinical study, Salgueiro et al (2015) analyzed the velopharyngeal (VP) activity of subjects with velopharyngeal dysfunction (VPD) by AR, as compared to rhinomanometry. A total of 41 adults, both genders, with repaired cleft palate, with or without a previously repaired cleft lip, and residual VPD on clinical assessment, without compensatory articulations for [p], [t], and [k] were included in this study. The outcome measures were as follows:

I. on AR, nasopharyngeal volumetric change (ΔV) during [p], [t], and [k], relatively to rest condition (decreases by less than 3 cm3 considered as absence of VP activity);
II. on modified anterior rhinomanometry, VP orifice area (areas greater than or equal to 0.05 cm2 considered as inadequate closure).

I. on AR, nasopharyngeal volumetric change (ΔV) during [p], [t], and [k], relatively to rest condition (decreases by less than 3 cm3 considered as absence of VP activity);
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The plosive [p] was used when comparing the techniques (n = 24). A mean ΔV decrease of 18 % was observed during [k], which was significantly lower ($p < 0.05$) than the decrease reported for individuals without VPD (30 %); ΔV values suggesting VPD were observed in 59 % subjects. Similar results were obtained for [p] and [t], which shall be used as stimulus, given that they did not involve the use of the tongue to lift the velum during VP closure, differently from the velar plosive [k]. Inadequate closure was seen in 85 % subjects. No correlation was observed between ΔV and VP orifice area. Agreement between techniques was observed in 51 % cases. The authors concluded that AR had low accuracy as a diagnostic method of VPD when compared to the gold standard method. Nevertheless, the technique showed potential as a method for monitoring the outcomes of clinical and surgical treatment of VPD aimed at increasing velar and pharyngeal activity.

In a retrospective, individual cohort study, Hsu and colleagues (2016) evaluated the effectiveness of septoplasty and the correlation between the subjective evaluations of a VAS and the NOSE questionnaire and active anterior rhinomanometry of the nasal airway after septoplasty. A total of 50 patients with chronic nasal obstruction were enrolled in the study. All 50 patients underwent septoplasty because of nasal septal deviation. Another 28 patients without nasal symptoms served as controls; VAS, NOSE, and active anterior rhinomanometry were used to measure the sensation of nasal obstruction. All measurements were performed in both groups pre-operatively and then repeated on 3 post-operative visits (3, 6, and 12 months). The mean

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VAS score, NOSE score, and the nasal resistance in the narrow side of the nose in the study group showed reduced symptoms at 3, 6, and 12 months post-operatively compared with the respective pre-operative measurements ($p < 0.001$, all). The VAS and NOSE scores did not significantly correlate with total nasal resistance pre-operatively or post-operatively. The VAS and nasal resistance in the obstructed nasal cavity correlated significantly pre-operatively ($p < 0.05$) but not post-operatively. The authors concluded that the subjective and objective symptoms of nasal obstruction had improved 1 year after septoplasty. A significant correlation between VAS scores and nasal resistance in the narrow side of the nose was found before surgery. However, the subjective and objective measurements of nasal obstruction lacked significant correlation post-operatively.

Fedok and colleagues (2016) stated that the middle vault of the nose continues to be a topic of interest among surgeons interested in aesthetic and functional rhinoplasty. These investigators presented currently accepted concepts regarding the significance of the middle vault of the nose in rhinoplasty and reviewed the more frequently advocated methods to be used in the correction of deficiencies. Spreader grafts may be at least as effective as flaring sutures in improving the airway. Studies have shown an improvement in quality of life and nasal breathing with the use of auto-spreader flaps. The correlation between AR and clinical symptoms of nasal obstruction, however, has fallen short of providing clear diagnostic value. The diagnosis of middle vault collapse and nasal valve obstruction remains largely clinical. The patient's reported symptoms of nasal obstruction were diagnostically considered along with the findings of clinical examination, including the findings of a modified Cottle maneuver. The use of spreader grafts and auto-spreader flaps has been popularized to correct problems in the middle vault of the nose and will be presented in detail in this manuscript.

Umihanic and co-workers (2016) noted that surgical and medical treatments of nasal obstruction are a common parts of otolaryngologist practice. The definitive treatment of deviated nasal septum is septoplasty. These investigators evaluated the values of subjective parameters, and active anterior rhinomanometry (AAR) parameters prior and 3 months after the septoplasty. They analyzed the subjective parameters ("NOSE" scale), the AAR parameters according to International Committee on Standardization of Rhinomanometry, on 40 patients; 30 healthy adult volunteers served as controls. None of the patients or healthy volunteers had previous history of nasal surgery or active rhinological disease. The post-operative improvement in symptoms of nasal obstruction obtained in 92.5 % patients and improvement parameters of the AAR in 42.5 % patients. The authors concluded that the correlation between the findings with rhinomanometry and subjective sensation of nasal patency remains uncertain. There still appeared to be only a limited argument for the use of rhinomanometry for quantifying surgical results. They stated that 3 months post-operative findings were very early results to interpret the permanent effects.

Chen and colleagues (2016) clarified the relationship between rhinomanometry measurements, fractional exhaled nitric oxide (FeNO), and spirometric measurements in asthmatic children. Patients' inclusion criteria: were age between 5 and 18 years, history of asthma with nasal symptoms, and no anatomical deformities. All subjects underwent rhinomanometric evaluations and pulmonary function and FeNO tests. A total of 84 children were enrolled. By rhinomanometry, the degree of nasal obstruction was characterized as follows:

I. no obstruction in 33 children,
II. slight obstruction in 29 children, and
III. moderate obstruction in 22 children;

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II. slight obstruction in 29 children, and
III. moderate obstruction in 22 children;

FeNO was significantly lower in patients without obstruction than those with slight or moderate obstruction. Dividing patients according to ATS Clinical Practice Guidelines regarding FeNO, patients less than 12 years with FeNO greater than 20 ppb had a lower total nasal airflow rate than those with FeNO less than 20 ppb. Patients greater than or equal to 12 years with FeNO greater than 25 ppb had a lower total nasal airflow rate than those with FeNO less than 25 ppb. The authors concluded that higher FeNO was associated with a

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lower nasal airflow and higher nasal resistance. They noted that these findings supported a relationship between upper and lower airway inflammation, as assessed by rhinomanometry and FeNO; and the results suggested that rhinomanometry may be integrated as part of the functional assessment of asthma. The authors stated that the present study provided preliminary results regarding the relationship between the upper and lower airways. As abnormalities in nasal patency are often associated with respiratory symptoms in pediatric patients, information on the degree of nasal patency is therefore useful in selecting decongestive, anti-allergic, anti-infectious, anti-inflammatory, and other therapies, and may help in the management of asthmatic children.

The main drawback of this study was the number of cases, which was simply too small to determine the full spectrum of relationships among these parameters. Larger numbers of cases in prospective, randomized studies are needed to determine relationships between rhinomanometric and spirometric measurements, IgE, allergic rhinitis symptom scores, and FeNO.

Sakai and associates (2018) stated that to provide clinical information and diagnosis in mouth breathers with transverse maxillary deficiency with posterior cross-bite, numerous examinations can be performed; but the correlation among these examinations remains unclear. In a cross-sectional study, these researchers evaluated the correlation between AR, computed rhinomanometry, and cone-beam CT in mouth breathers with transverse maxillary deficiency. This study was conducted in 30 mouth breathers with transverse maxillary deficiency (age of 7 to 13 years) patients with posterior cross-bite. The examinations assessed:

I. AR: nasal volumes (0 to 5 cm and 2 to 5 cm) and minimum cross-sectional areas 1 and 2 of nasal cavity;
II. computed rhinomanometry: flow and average inspiratory and expiratory resistance; and
III. cone-beam CT: coronal section on the head of inferior turbinate (widths 1 and 2), middle turbinate (widths 3 and 4) and maxilla levels (width 5);

I. AR: nasal volumes (0 to 5 cm and 2 to 5 cm) and minimum cross-sectional areas 1 and 2 of nasal cavity;
II. computed rhinomanometry: flow and average inspiratory and expiratory resistance; and
III. cone-beam CT: coronal section on the head of inferior turbinate (widths 1 and 2), middle turbinate (widths 3 and 4) and maxilla levels (width 5);

AR and computed rhinomanometry were evaluated before and after administration of vasoconstrictor. Results were compared by Spearman's correlation and Mann-Whitney tests ($\alpha = 0.05$). Positive correlation was observed between:

I. flow evaluated before administration of vasoconstrictor and width 4 (Rho = 0.380) and width 5 (Rho = 0.371);
II. width 2 and minimum cross-sectional areas 1 evaluated before administration of vasoconstrictor (Rho = 0.380);
III. flow evaluated before administration of vasoconstrictor and nasal volumes of 0 to 5 cm (Rho = 0.421), nasal volumes of 2 to 5 cm (Rho = 0.393) and minimum cross-sectional areas 1 (Rho = 0.375);
IV. width 4 and nasal volumes of 0 to 5 cm evaluated before administration of vasoconstrictor (Rho = 0.376), nasal volumes of 2 to 5 cm evaluated before administration of vasoconstrictor (Rho = 0.376), minimum cross-sectional areas 1 evaluated before administration of vasoconstrictor (Rho = 0.410) and minimum cross-sectional areas 1 after administration of vasoconstrictor (Rho = 0.426);
V. width 5 and width 1 (Rho = 0.542), width 2 (Rho = 0.411), and width 4 (Rho = 0.429).

I. flow evaluated before administration of vasoconstrictor and width 4 (Rho = 0.380) and width 5 (Rho = 0.371);
II. width 2 and minimum cross-sectional areas 1 evaluated before administration of vasoconstrictor (Rho = 0.380);
III. flow evaluated before administration of vasoconstrictor and nasal volumes of 0 to 5 cm (Rho = 0.421), nasal volumes of 2 to 5 cm (Rho = 0.393) and minimum cross-sectional areas 1 (Rho = 0.375);

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IV. width 4 and nasal volumes of 0 to 5 cm evaluated before administration of vasoconstrictor (Rho = 0.376), nasal volumes of 2 to 5 cm evaluated before administration of vasoconstrictor (Rho = 0.376), minimum cross-sectional areas 1 evaluated before administration of vasoconstrictor (Rho = 0.410) and minimum cross-sectional areas 1 after administration of vasoconstrictor (Rho = 0.426);
V. width 5 and width 1 (Rho = 0.542), width 2 (Rho = 0.411), and width 4 (Rho = 0.429).

Negative correlation was observed between:

I. width 4 and average inspiratory resistance (Rho = -0.385);
II. average inspiratory resistance evaluated before administration of vasoconstrictor and nasal volumes of 0 to 5 cm (Rho = -0.382), and average expiratory resistance evaluated before administration of vasoconstrictor and minimum cross-sectional areas 1 (Rho = -0.362).

I. width 4 and average inspiratory resistance (Rho = -0.385);
II. average inspiratory resistance evaluated before administration of vasoconstrictor and nasal volumes of 0 to 5 cm (Rho = -0.382), and average expiratory resistance evaluated before administration of vasoconstrictor and minimum cross-sectional areas 1 (Rho = -0.362).

The authors concluded that there was correlation between AR, computed rhinomanometry, and cone-beam CT in mouth breathers with transverse maxillary deficiency. These investigators stated that the findings of this study have a special relevance for future research challenges. They noted that in the future, comprehensive studies should be carried out with larger sample sizes and include comparisons between the groups mentioned in the drawbacks of this study, as well as the results obtained from the long-term treatment of transverse maxillary deficiency with maxillary expansion.

The drawbacks of this study included:

I. small sample size (n = 30),
II. non-inclusion of healthy controls for correlation between the tests, and
III. non-inclusion of controls with deficiency and nasal breathers.

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II. non-inclusion of healthy controls for correlation between the tests, and
III. non-inclusion of controls with deficiency and nasal breathers.

Controls were not included due to the need of cone-beam CT examinations.

Shohara and colleagues (2017) stated that numerous techniques have been used to reduce epistaxis during naso-tracheal intubation. Rhinometry can assess nasal patency in pre-operative conditions. However, the possible role of rhinometry in routine naso-tracheal intubation has not been studied. In this study, a total of 101 patients undergoing dental and maxilla-facial surgery that required general anesthesia and naso-tracheal intubation were enrolled. These researchers examined if symmetry or any asymmetry in bilateral airflow patterns by condensation of the expiration, assessed by pre-operative rhinometry on seated position, increased the incidence of epistaxis and the need for a naso-gastric catheter to guide the endo-tracheal tube into the oropharynx. They also compared the incidence of changing the site of nasal intubation between the assessment by rhinometry and by cone-beam CT analysis of nasal airspace in the inferior meatus. Patients with any asymmetry in bilateral airflow patterns were 18 % (n = 18), the remaining 82 % (n = 83) had symmetric bilateral nasal cavities. Patients with any asymmetry were more likely to need a guiding naso-gastric catheter than patients with symmetry (22 % versus 3.6 %, p = 0.018). The incidence of epistaxis was higher in patients with any asymmetry (39 %) than those with symmetry (16 %), but there was no significant difference between groups (p = 0.055). The site of intubation was changed more frequently based on cone-beam CT analysis than by rhinometry (38 % versus 11 %, p = 0.043). The authors concluded that pre-operative rhinometry may be a valuable objective tool to evaluate nasal patency for naso-tracheal intubation in patients who undergo dental and maxilla-facial surgery.

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Bock and colleagues (2017) stated that cystic fibrosis (CF) patients almost regularly reveal sino-nasal pathology. These researchers evaluated association between objective and subjective measurements of sino-nasal involvement comparing nasal airflow obtained by AAR, nasal endoscopic findings, and symptoms assessed with the Sino-Nasal Outcome Test-20 (SNOT-20). Nasal cavities were explored by anterior rigid rhinoscopy and findings were compared to inspiratory nasal airflow measured by AAR to quantify nasal patency and subjective health-related quality of life (QOL) in sino-nasal disease obtained with the SNOT-20 questionnaire. Relations to upper and lower airway colonization with Pseudomonas aeruginosa, medical treatment, and sino-nasal surgery were analyzed. A total of 124 CF patients were enrolled (mean age of 19.9 ± 10.4 years, range of 4 to 65 years). A significant association of detection of nasal polyposis (NP) in rhinoscopy was found with increased primary nasal symptoms (PNS), which include "nasal obstruction", "sneezing", "runny nose", "thick nasal discharge", and "reduced sense of smell". At the same time patients with pathologically decreased airflow neither showed elevated SNOT-20 scores nor abnormal rhinoscopic findings. Altogether, rhinomanometric and rhinoscopic findings were not significantly related. The authors concluded that among SNOT-20 scores the PNS sub-score was related to rhinoscopically detected polyposis and sino-nasal secretion. Thus, these investigators recommend including short questions regarding PNS into CF-routine care. At the same time these findings showed that a high inspiratory airflow was not associated with a good sensation of nasal patency. They stated that rhinomanometry is not needed within routine CF-care, but it can be interesting as an outcome parameter within clinical trials.

Numminen and colleagues (2003) stated that in recent years increasing evidence has been provided on frequent simultaneous co-existence of inflammatory diseases and allergies in upper and lower airways. To achieve a good standard of measurement of upper airways, an objective method should be used. A total of 48 nasal cavities with nasal stuffiness associated with CRS were measured with AR and high-resolution CT volumetry (HRCTV). Comparison of volumes and minimum cross-sectional areas (CSAs) measured with these methods was performed. The volumes measured from the nostril with both methods were the anterior (0 to 10 mm), middle (11 to 40 mm) and posterior (41 to 70 mm) volumes. The AR CSA curve was analyzed based on 2 minimal notches corresponding to local minimal areas. A series of 1-mm coronal CT images without intervening gaps were made and analyzed based on 2 minimal voxel values, which were later converted to CSAs corresponding to local, minimum cross-sectional areas (MCA). Furthermore, the distances of these 2 MCAs from the nostril were also measured. Strong statistically significant ($p < 0.05$) correlations were found between AR and CT volumetry (CTV) volumes in the anterior ($r = 0.83$) and middle ($r = 0.77$) parts of the nasal cavity. In the posterior part of the nasal cavity a statistically significant ($p < 0.05$) correlation was also found ($r = 0.62$). Good agreements between the AR and CTV volumes in the anterior and middle parts of the nasal cavities were confirmed with Bland-Altman plots. Correlations among the MCAs were weaker ($r = 0.59$ and $r = 0.55$). The authors concluded that these findings suggested that the reliability of AR appeared sufficient for clinical and scientific use in the nasal cavities. Reliability was very good in the anterior and middle parts of the nasal cavities, while strong conclusions based on evaluation of the posterior part should be avoided due to decreasing accuracy. They noted that AR is a clinically reliable method for measuring nasal cavity geometry in the anterior and middle parts of the nasal cavity. Moreover, they stated that further research is needed before standardization of AR can be accomplished.

Nathan and colleagues (2005) stated that nasal obstruction can be monitored objectively by measurement of nasal airflow, as evaluated by nasal peak flow, or as airways resistance/conductance as evaluated by rhinomanometry. Peak flow can be measured during inspiration or expiration. Of these measurements, nasal inspiratory peak flow is the best validated technique for home monitoring in clinical trials. The equipment is portable, relatively inexpensive, and simple to use. One disadvantage, however, is that nasal inspiratory peak flow is influenced by lower airway as well as upper airway function. Rhinomanometry is a more sensitive technique that is specific for nasal measurements. The equipment, however, requires an operator, is more expensive, and is not portable. Thus, it is applicable only for clinic visit measures in clinical trials. Measurements require patient co-operation and co-ordination, and not all can achieve repeatable results. Thus, this objective measure is best-suited to laboratory challenge studies involving smaller numbers of selected volunteers. A non-physiological measure of nasal patency is AR. This sonic echo technique measures internal nasal luminal volume and the minimum cross-sectional area. The derivation of these measures from the reflected sound waves requires complex mathematical transformation and makes several theoretical assumptions. Despite this, however, such measures correlate well with the nasal physiological measures, and the nasal volume measures have been shown to relate well to results obtained by imaging

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techniques such as CT scanning or magnetic resonance imaging (MRI). Like rhinomanometry, AR is not suitable for home monitoring and can be applied only to clinic visit measures or for laboratory nasal challenge monitoring. It has advantages in being easy to use, in requiring little patient co-operation, and in providing repeatable results. In addition to nasal obstruction, allergic rhinitis is recognized to be associated with impaired muco-ciliary clearance and altered nasal responsiveness. Measures exist for the monitoring of these aspects of nasal dysfunction. Although measures of muco-ciliary clearance are simple to perform, they have a poor record of reproducibility. Their incorporation into clinical trials is thus questionable, although positive outcomes from therapeutic intervention have been reported. The authors concluded that measures of nasal responsiveness are at present largely confined to research studies examining disease mechanisms in allergic and non-allergic rhinitis. The techniques are insufficiently standardized to be applied to multi-center clinical trials; but could be used in limited-center studies to gain insight into the regulatory effects of different therapeutic modalities.

Maalouf and associates (2016) stated that nasal valve collapse is a dynamic abnormality that is currently diagnosed purely on the basis of clinical features and thus subject to certain interpretation. In an observational, prospective study, these researchers developed a new and reliable functional test to objectively characterize nasal valve collapse. This trial included consecutive patients referred to the authors' center for exploration of chronic nasal congestion. Participants were classified into 2 groups according to their symptoms and clinical abnormalities: the nasal valve collapse (NV+) group when nasal valve collapse was clinically detected during moderate forced inspiration and/or when the feeling of nasal congestion improved during passive nasal lateral cartilage abduction ($n = 32$); and the no-nasal valve collapse (NV−) group for the others ($n = 23$). All patients underwent nasal functional tests (posterior rhinomanometry and AR) before and after topical nasal decongestion. These investigators compared the difference between the pressure flow of the inspiratory and expiratory phases during posterior rhinomanometry [flow rate inspiratory-expiratory difference (FRIED) test] between the 2 groups. The difference between the absolute value of inspiratory and expiratory flow was significantly higher in the NV+ group than in the NV− group both before and after topical decongestion. The cut-off value for the FRIED test was −0.008 l/s with a good sensitivity (82 %) and a specificity of 59 %. The authors suggested that the FRIED test constituted an objective and easy-to-apply technique to diagnose nasal valve collapse in daily practice. Moreover, these researchers noted that measuring nasal compliance with AR is not a reliable way to characterize nasal valve collapse. Nevertheless, compliance measurements are of great interest when exploring nasal obstruction and could detect any potential dysfunction posterior to the valve. They stated that this study provided the 1st proof-of-principle of a useful, reliable, and easy-to-perform test to objectively and quantitatively characterize nasal valve collapse in patients complaining of nasal obstruction. In the future, the FRIED test could also be used to evaluate the efficacy of nasal valve treatments, both prosthetic and surgical.

Lai and associates (2017) noted that there is no standardized scheme for pre-operative evaluation of adenoid hypertrophy or a consensus on surgical indications for adenoidectomy in children with otitis media with effusion (OME), especially for young children intolerant to nasal endoscopic assessment. These investigators evaluated the efficacy and reliability of AR in evaluating benefits from adenoidectomy in children with OME. Children with OME who were scheduled for surgical intervention were reviewed and AR tests performed pre-operatively and post-operatively. Subjects were divided into 2 groups based on the surgical strategy (Group I: tympanostomy tube placement alone; Group II: tympanostomy tube placement plus adenoidectomy). Correlation and regression analyses were performed to assess the relationship between findings of AR and nasal endoscopy; AR parameters including minimal nasal cross-sectional area (MCSA), and naso-pharyngeal volume (NPV), as well as scores of subjective symptoms were obtained to evaluate the utility of AR pre- and post-surgery. A total of 65 children aged 4 to 10 years who met the inclusion criteria were included. No significant differences in gender or age distribution were observed between Group I and Group II; MCSA, as well as NPV significantly decreased in Group II when compared with Group I ($p = 0.000$). A significant inverse correlation was observed between NPV and choanal obstruction ratio in both groups I ($r = -0.625$, $p < 0.001$) and II ($r = -0.570$, $p < 0.001$). A significant difference between pre-operative and post-operative NPV and subjective symptom scores was observed in group II after adenoidectomy ($p = 0.000$). The authors concluded that AR parameters showed a good clinical correlation with findings of nasal endoscopy and thus may be useful for evaluating candidacy for surgical adenoidectomy among children with OME, especially in whom pre-operative nasal endoscopic examination was not feasible. Additionally, AR can reveal the changes occurring within the naso-pharyngeal passage before and after adenoidectomy.

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Wartelle and colleagues (2018) stated that the acoustic reflection method (ARM) is a non-invasive technique that utilizes the reflection of acoustic waves to measure the cross-sectional area (CSA)of nasal cavities in adults and patency of endotracheal tubes. Characteristics and volume of normal nasal cavities in pre-school children has so far not been studied. In a prospective, mono-centric study, these researchers determined the optimal ARM recording and the MCSA and NPV values in healthy pre-school children (less than 6 years of age). A total of 70 children (aged 2 to 5 years) were included in the study. Reliable measures were difficult to obtain in children younger than 2 years of age. The use of a standard nose-piece and a single-use surgical filter enabled reliable, serial recordings. Mean MCSA values were 0.46, 0.53 and 0.58 cm2 in the 24 to 35, 36 to 47 and 48 to 60 months-old age groups, respectively. Mean NPV values were 2.14, 2.59, and 2.86 cm3 in the same age groups. The MCSA and NV values were significantly correlated with height, age and weight. The authors concluded that the ARM was feasible in children over the age of 2 and appeared to be a promising non-invasive tool to study the nasal cavity patency, anatomy, and volume.

Furthermore, an UpToDate review on "Clinical presentation, diagnosis, and treatment of nasal obstruction" (Bhattacharyya, 2018) states that "Several other tests can be performed to help characterize nasal obstruction. The data supporting the use of these measurements are somewhat controversial and results can be less than definitive. Thus, these tests are usually ordered under select clinical situations after specialist evaluation … Acoustic rhinometry is a simple, noninvasive measure of cross-sectional area of the nasal cavity longitudinally along the nasal passageway … The degree of nasal obstruction, as measured objectively by acoustic rhinometry, peak nasal airflow, or rhinomanometry, may not correlate well with the patient's subjective sense of nasal obstruction".

Aksoy and colleagues (2018) noted that seasonal allergic rhinitis (SAR) is common in children and hyposmia is a major symptom affecting the QOL. These investigators evaluated olfactory dysfunction in pediatric patients with SAR and correlated the findings with AR measurements. A total of 40 children, diagnosed as moderate and severe SAR based on clinical findings, ARIA (Allergic rhinitis and its impact on asthma) classification and prick test results were enrolled in the study. Endoscopic nasal examination, AR, total nasal symptom score (TNSS) and Connecticut Chemosensory Clinical Research Center (CCCRC) tests were performed "in season" (May to August) and "out season" (November to February); 3 patients who did not show up in "out season" examinations were excluded from the study. The ages of the children ranged between 8 and 18 years with a hyposmia increased and odor identification decreased ($p<0.005$, $p=0.003$, respectively), whereas no differences were found between odor thresholds and the discrimination values ($p>0.05$). Mean CCCRC value was obstruction score ($r=-0.340$, $p=0.04$), subjective hyposmia ($r=-0.44$, $p=0.007$) and TNSS ($r=-0.494$, $p=0.02$). Although some of the AR parameters were lower during allergy season, there were no correlations between AR parameters and CCCRS values. The authors concluded that nearly 50 % of the children with AR reported a mild-to-moderate hyposmia during pollen season and there was a decrease in odor identification, which could be easily shown using a CCCRC test.

In a prospective study, Distinguin and colleagues (2019) compared nasal volume and MCA between 3 groups of children: "achondroplasia", "Down syndrome" and "control". The control group corresponded to children with suspicion of sleep disorder disease and without cranio-facial malformation. The second objective was to correlate AR measurements with the obstructive apnea-hypopnea index (OAHI). This trial was carried out between February and July 2017, in a tertiary-care center. The following data were collected: demographic characteristics, medical and surgical history, nasal volume, MCA, and OAHI. A total of 83 children were included. The mean nasal volume was lower in achondroplasia group compared to control group: 2.75 cm3 versus 3.60 cm3 ($p = 0.02$, 95 % confidence interval [CI]: 0.0694 to 0.7456). Negative correlation was found between the nasal volume and the OAHI for children with achondroplasia ($T = -0.37$; $p = 0.02$). The authors concluded that AR was an effective tool for evaluating nasal obstruction in children; and nasal obstruction was correlated to OAHI in achondroplasia. These researchers stated that AR could become a routine tool in the management of nasal obstruction of children with cranio-facial malformations.

Furthermore, an UpToDate review on "Nasal obstruction: Diagnosis and management" (Bhattacharyya, 2023) states that "The degree of nasal obstruction, as measured objectively by acoustic rhinometry, peak nasal airflow, or rhinomanometry, may not correlate with a patient's subjective degree of nasal obstruction. As an example, minimal changes in nasal patency (measured objectively) may be experienced as a substantial problem for an individual patient … Several other tests can be performed to help

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characterize nasal obstruction. The data supporting the use of these measurements are somewhat controversial and results can be less than definitive. Thus, these tests are usually ordered under select clinical situations after specialist evaluation"; and these tests include AR and rhinomanometry.

Araujo-Martins and co-workers (2020) stated that although the pathophysiology of nasal polyposis is incompletely understood, rhinologists have seldom studied it with rhinomanometry or PNIF due to technical limitations and the perception that polyp size might impair reproducibility and the usefulness of recordings. These researchers examined how measures of rhinomanometry and PNIF relate to disease activity. A total of 19 patients with polyps, 15 patients with chronic sinusitis without polyps and 11 negative controls were examined with AAR and PNIF. Sinusitis and polyp patients were re-evaluated following medical treatment. Polyp patients had the highest median Lund-Mackay score (14) and a median Johansen score of 1. PNIF and its variation following treatment were also lowest in this group (median 90 L/min before and following treatment, median variation of 0 L/min). Nasal resistance was similar between groups, and only correlated with Johansen score (Spearman = 0.517, p = 0.048) following treatment. The authors concluded that he findings of this study suggested that evaluating polyp patients using rhinomanometry and PNIF may provide useful and reproducible data. Several findings considered together suggested that polyp size is not the main determinant of nasal functional changes in these patients, warranting further studies to examine if PNIF changes reflect sinus inflammation or merely airway obstruction. Moreover, these researchers stated that it appeared differences between sinusitis patients with or without polyps may be naturally small and PNIF may be a more useful clinical biomarker of disease severity. The findings of this study should encourage researchers to continue investigating the clinical usefulness of PNIF and rhinomanometry, as well as their contribution to the study of NP pathophysiology.

Brindisi and colleagues (2021) noted that allergic rhinitis (AR) and asthma are 2 common atopic diseases, often associated with a common ethiopathogenesis characterized by a Th2 inflammatory response with the release of many biomarkers, such as nitric oxide (NO). These researchers compared inflammatory (nasal NO [nFeNO] and exhaled NO [eFeNO]) and functional (mean nasal flow [mNF] and forced expiratory volume in the 1st second [FEV1]) parameters in AR children with or without asthma in comparison to controls. Secondly, these investigators identified nFeNO cut-off values and verified their reliability to predict the presence of rhinitis or asthma alone or in combination. They enrolled 160 children (6 to 12 years of age) with AR and/or asthma divided into 4 groups: controls, AR, asthma, and AR + asthma. All children underwent the following inflammatory and functional measurements: nFeNO, eFeNO, mNF and FEV1. These researchers observed that levels of nFeNO were extremely higher in children with AR and even more in those with AR + asthma in respect to controls. Notably, all the pathological conditions, especially AR + asthma, showed significantly lower values of mNF compared to healthy children. A negative correlation linked mNF and nFeNO. Then, these investigators found eFeNO values significantly higher in all the pathological groups compared to controls, with major values of this marker in patients affected by asthma and AR + asthma, as well as FEV1 values significantly lower in all the disease groups, especially in children with asthma and AR+ asthma. ROC curve analysis showed that nFeNO was a great predictor for rhinitis alone or with asthma, revealing an accurate cut-off of 662 ppb. The authors concluded that nFeNO measurement was non-invasive, easy to perform, economic and a valuable test in case of AR alone or in association with asthma. Therefore, it should be used in patients with rhinitis, together with AAR to diagnose and estimate the degree of nasal obstruction but also in children with asthma to evaluate their nasal involvement and improve the therapeutic management.

The authors stated that this study had several drawbacks. First, the sample size for each group was quite small and subjects were all enrolled from the same out-patient setting; thus, further studies are needed to confirm these findings. Moreover, the nFeNO technique of measurement is not yet standardized, it needs methodological improvements to become a reliable tool that is comparable among different studies.

Calvo-Henriquez and associates (2021) stated that the main causes for objectively confirmed chronic impaired nasal breathing in children are adenoid and turbinate hypertrophy. Turbinate hypertrophy may be addressed by turbinate surgery; however, specialized guidelines include no specific indications for pediatric patients. The decongestant test consists of simulating the effect of turbinate surgery by means of a nasal decongestant. This project, developed by the YO-IFOS rhinology group, aimed to establish a cut-off value for the nasal decongestant test with rhinomanometry to select children for turbinate surgery. Children

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between 4 and 15 years of age were included. Cases were consecutively selected from children affected by turbinate hypertrophy undergoing turbinate radiofrequency ablation (RFA) with or without adenoidectomy. Controls were consecutively selected from a sample of healthy children. All the subjects were examined with AAR with and without nasal decongestant. Participants included 72 cases and 24 healthy controls. There was a statistically significant difference in the improvement with the decongestant between cases (57.91 %) and controls (15.67 %). The ROC curve revealed an area under the curve of 0.97. The highest amount of correctly classified individuals (93.44 %) corresponded to the cut-off value of 31.66 %. However, the value with the highest specificity and highest Youden's index was the 38.88 % improvement in nasal resistance with nasal decongestant. The authors concluded that a preliminary cut-off value for the decongestant test used with rhinomanometry in children has been established. These researchers stated that this test could help identify children for turbinate surgery.

Ta and colleagues (2021) noted that common sino-nasal disorders include CRS, AR, and a deviated nasal septum (DNS), which often co-exist with shared common symptoms including nasal obstruction, olfactory dysfunction, and rhinorrhea. Various objective outcome measures and patient-reported outcome measures (PROMs) are used to examine disease severity; however, there is limited evidence in the literature on the correlation between them. In a systematic review, these investigators examined the relationship between them and provided recommendations. They carried out a search of Medline and Embase; and identified studies quantifying correlations between objective outcome measures and PROMs for the sino-nasal conditions using a narrative synthesis. A total of 59 studies met inclusion criteria. For nasal obstruction, rhinomanometry showed a lack of correlation whereas PNIF showed the strongest correlation with PROMs ($r > 0.5$). The Sniffin' Stick test showed a stronger correlation with PROMs ($r > 0.5$) than the University of Pennsylvania Smell Identification Test (UPSIT) ($r < 0.5$); and CT sinus scores showed little evidence of correlation with PROMs and nasal endoscopic ratings (weak correlation, $r < 0.5$). The authors concluded that objective outcome measures and PROMs evaluating sino-nasal symptoms were poorly correlated, and they recommended that objective outcome measures be used with validated PROMs depending on the setting. PNIF should be used in routine clinical practice for nasal obstruction; rhinomanometry and acoustic rhinometry may be useful in research. The Sniffin' Sticks test is recommended for olfactory dysfunction with UPSIT as an alternative. CT scores should be excluded as a routine CRS outcome measure, and endoscopic scores should be used in combination with PROMs until further research is carried out.

Hassegawa and associates (2021) compared the nasal cavity geometry of children and teenagers with cleft lip and palate and maxillary atresia by 2 methods: cone-beam CT, considered the gold standard, and acoustic rhinometry. Data on cone-beam CT and acoustic rhinometry examinations of 17 children and teenagers with cleft lip and palate and maxillary atresia, previously obtained for orthodontic planning purposes, were evaluated prospectively. Using Dolphin Imaging 11.8 software, the nasal cavity was reconstructed by 2 evaluators, and the internal nasal volumes were obtained. Using rhinometry, the volumes of regions V1 and V2 were measured. The values of each examination were then compared at a significance level of 5 %. Statistical analysis showed high intra- and inter-rater reproducibility in the cone-beam CT analysis. The mean internal nasal volumes (± standard deviation) obtained using acoustic rhinometry and cone-beam CT corresponded to 6.6 ± 1.9 cm3 and 8.1 ± 1.5 cm3, respectively. The difference between the examinations was 17.7 %, which was considered statistically significant ($p = 0.006$). The authors concluded that nasal volumes measured by the 2 methods were different, presenting discrepancies in the measurements. The cone-beam CT (gold standard technique) identified larger volumes than acoustic rhinometry in the nasal cavity. These researchers stated that determining which test reflects clinical reality is an essential future step.

Valtonen et al (2022) noted that AR is widely used in the examination of patients with nasal congestion; however, it only has a partial correlation with patient's symptoms. The use of cone beam computed tomography (CBCT) scans are mainly on the paranasal sinuses and less on the nasal cavities; thus, information acquired from CBCT scans is not used to its full extent. These investigators examined patients with enlarged inferior turbinates; and compared the use of 3D volumetric measurements and CSA measurements taken from CBCT scans to results obtained from AR. A total of 25 patients with enlarged inferior turbinates were enrolled in this trial. CBCT scans were obtained pre-operatively and at 12 months post-operatively. 3D volumetric and CSA measurements were compared to results from AR, the VAS and Glasgow Health Status Inventory (GHSI) questionnaires. A statistically significant change in 3D volume and CSA was measured in the anterior part of the inferior turbinate and surrounding air space after inferior

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turbinate surgery. VAS and GHSI results had mild correlations with the 3D volume and CSA measurements of the anterior part of the inferior turbinate; and AR correlated with the air space 3D volume measurements in the anterior part. The authors concluded that fully utilized CBCT scans provided more comprehensive and accurate information. In addition, 3D analysis of the inferior turbinates provided valuable information and more precise measurements compared to AR.

Gagnieur et al (2022) stated that internal valve collapse is a frequent cause of nasal obstruction but remains poorly understood and is sometimes treated inappropriately as a result. No functional or imaging test for the condition has been validated and the reference diagnostic technique is physical examination. In a diagnostic accuracy case-control study, these researchers examined the potential of 4-phase rhinomanometry as a diagnostic test for internal valve collapse. In this trial, the nostrils of adult patients consulting for chronic nasal obstruction were classified as "collapsed" or "non-collapsed" based on clinical findings; 4-phase rhinomanometry was carried out in all patients. The area defined by the path of the flow/pressure curve in the 2 phases of inspiration (the "inspiratory loop area" or "hysteresis loop area") was calculated for both nasal cavities and the threshold value with the highest Youden index was identified. A total of 66 patients (132 nostrils) were included with 72 nostrils classified as collapsed and 60 as non-collapsed. Before nasal decongestion, the inspiratory loop area with the highest Youden index was 17.3 Pa L s-1 and the corresponding sensitivity and specificity were 88.3 % (95 % CI: 80.0 % to 95.0 %) and 89.9 % (82.6 % to 95.7 %), respectively. The authors concluded that in these patients, a cut-off inspiratory loop area in 4-phase rhinomanometry data reproduced clinical diagnoses of internal valve collapse with high sensitivity and specificity. These researchers stated that this method may offer a firmer basis for treatment indications than subjective physical examinations. Moreover, these investigators stated that larger studies with a pre-defined threshold loop area are needed to confirm these promising results. Level of Evidence = IV.

The authors stated that the drawbacks of this study included its small size (n = 66 subjects) and the fact that the 4-phase rhinomanometry measurements were carried out on each nostril separately, even if the contralateral nostril was occluded with medical tape rather than a nasal plug to avoid altering the structure and biomechanical properties of the studied nostril.

Calvo-Henriquez et al (2022) stated that nasal obstruction is a common complaint in pediatric otolaryngology. There are several concerns regarding how nasal obstruction should be measured. This debate is even more important in children, as they could experience difficulties in being sensitive to their symptoms or even expressing them. In an observational, cross-sectional study, these researchers examined the ability of children (and their parents) to evaluate their nasal obstruction. A total of 4 cohorts of children were consecutively selected from a 3rd level referral Hospital. Cohort A (children suffering solely turbinate enlargement), B (adenoid enlargement only), C (adenoid and turbinate enlargement), while cohort D were healthy controls. Children and parents were asked to rate nasal patency via a Likert scale from 0 (no patency, complete obstruction of the nose) to 10 (complete patency, it is easy to breathe through the nose). All subjects underwent rhinomanometry. Results of nasal resistance were relativized according to pediatric reference values per each age subgroup. A total of 146 subjects were included -- Cohort A (n = 54), B (n = 40), C (n = 28), D (n = 24). There was a poor but significant correlation between parents' assessment and nasal resistance (rho = -0.28; p = 0.004). In children, there was no significant correlation with nasal resistance (rho = -0.14; p = 0.17). Stratified by severity, only children (and their parents) with good nasal breathing showed good correlation values with the VAS. Stratified by age, the correlation was only significant for parents of children 12 years of age or older. The authors concluded that this study has demonstrated a good ability to rate nasal patency by healthy children and their parents, but a poor ability for children suffering from impaired nasal breathing. These researchers suggested combining subjective assessment of nasal patency with objective measurements such as rhinomanometry in children. This was an observational, cross-sectional study; its findings need to be validated by well-designed studies.

AlEnazi et al (2023) stated that accurate methods are needed to examine the anatomy of the internal nasal valve (INV); however, there is currently no ideal measurement technique. In a systematic review, these investigators aimed to establish a comprehensive INV assessment tool, compared different INV diagnostic tools, and established the most ideal measurement technique for the evaluation of the INV. This systematic review and meta-analysis were carried out according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. These researchers conducted a systematic search in

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PubMed, Medline, the Cochrane Library (Cochrane Databases of Systematic Reviews), and the Cochrane Register of Controlled Trials (CENTRAL) for studies examining INV that were performed between 1996 and 2023. Of the 421 total database searches, 23 studies were found, covering a total of 974 participants (6 studies examined the accuracy of different diagnostic methods, with 2 of these studies comparing 2 different diagnostic modalities, and 17 studies measured INV angle). Based on the STROBE tool for quality appraisal the mean score was 16.92 ± ± 2.29, indicating a moderate quality. When comparing INV angle values from pre-operative and post-operative records as obtained from CT readings, results showed no significant differences between the pre- and post-operative values (mean difference [MD] = -1.8, 95 % CI: -4.8 to 1.2, p = 0.227). The authors concluded that an ideal measurement modality for evaluation of the INV has not been established. Acoustic rhinometry has the highest accuracy, followed by rhinomanometry, then CT scan then endoscopy. Meta-analysis revealed no significant differences between the pre- and post-operative INV angle values. Furthermore, the findings of this meta-analysis indicated a significant heterogeneity across studies, explained by different surgical interventions and different techniques between facial plastic surgeons and aesthetic plastic surgeons. A purported angle of 10 to 15 degrees exists between the upper lateral cartilage and the septum in patients without nasal obstruction, which may perhaps account for the variation among different diagnostic modalities; and the accuracy of the re-formatted form of CT plane scan is not well-established yet. Thus, these researchers recommended more studies to establish and test the accuracy of the new format of diagnostic modality in evaluating INV. Furthermore, the cut-point of the INV angle should be established for each diagnostic modality.

The authors stated that this was the 1st comprehensive review comparing multiple tools and modalities to establish valuable insights into the accurate assessment of choice in INV measurements. However, several drawbacks need to be addressed including the majority of the included studies had a small number of patients, and some did not accurately describe the study design. Additionally, the STROBE check-list mean score of the included studies was indicative of a moderate quality.

Sunnergren et al (2023) noted that AAR is often used in Swedish routine clinical practice to decide if septoplasty is necessary; however, the scientific basis for the method needs to be strengthened. In a prospective, longitudinal study, these researchers examined NAR, paradoxical reactions to pharmacological decongestion, and test-retest characteristics of the Rhino-Comp AAR in healthy volunteers. AAR was conducted before and after decongestion at baseline and after 6 months or longer on 60 healthy volunteers. The relationships between NAR, height, weight, BMI, sex, and allergic rhinitis were examined by regression analyses. Descriptive statistics were used to evaluate paradoxical reactions. Test-retest and repeatability characteristics were evaluated with intra-class coefficients (ICC), Cronbach's α, and standard error (SE) of measurement. No statistically significant differences were observed between genders or nasal cavity sides. NAR was statistically significantly related to height. Short- and long-term test-retest characteristics were good with ICC and Cronbach's α greater than 0.75. The minimal significant difference in NAR log10V2 values between the 2 measurements was 0.11 and 0.09 (long-term and short-term). Paradoxical reactions to pharmacological decongestion were rare, mostly weak, and not evidently reproducible. The authors concluded that AAR NAR values were related to height but not to sex or side, findings that must be taken into consideration when AAR is used in clinical practice. The method of pharmacological decongestion appeared to be superior to methods used in previous studies and these investigators recommended that their method should be used in future studies as well as clinical practice. For the 1st time, the minimal difference between 2 measurements that with confidence signified a true change was established, a finding with great importance for the evaluation of the outcome of septal surgery. Paradoxical reactions to pharmacological decongestion existed but the best definition remains to be decided and the clinical significance of paradoxical reactions needs to be further studied, especially in subjects with nasal stenosis. Moreover, these researchers stated that these findings need to be corroborated in a multi-center study with a larger population including both healthy subjects as well as subjects with different degrees of nasal septum deviations. In addition, intra- and inter-rater studies of Rhino-Comp AAR measurements are needed. Paradoxical reactions to pharmacologic decongestion also need to be further studied so that an appropriate characterization and definition of the reaction can be established. Correlations between subjective assessments of nasal function and AAR data both before and after surgery in septoplasty patients, as well as the effect of septoplasty on AAR results, need to be studied before any evidence-based guidelines for the use of AAR in clinical practice can be made.

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Fujito et al (2023) stated that nasal breathing disorders are associated with OSA syndrome and influence the availability of continuous positive airway pressure (CPAP) therapy; however, information is scarce regarding the impact of nasal resistance assessed by rhinomanometry on CPAP therapy. In a retrospective, single-center study, these investigators examined the relationship between CPAP adherence and nasal resistance assessed by rhinomanometry, and identified clinical findings that could affect adherence to CPAP therapy for patients with OSA. This study included 260 patients (199 men, 61 women; age of 58 years [inter-quartile ranges (IQR) 50 to 66]) with a new diagnosis of OSA who underwent rhinomanometry (before, and 1 and 3 months after CPAP introduction) between January 2011 and December 2018. CPAP use was recorded, and the good and poor CPAP adherence groups at the time of patient registration were compared. In addition, those with improved and unimproved pre-CPAP high rhinomanometry values were also compared. Their AHI by PSG at diagnosis was 45.6 (IQR 33.7 to 61.6)/hour; however, the residual respiratory event (estimated AHI) at enrollment was 2.5 (IQR 1.4 to 3.9)/hour, and the usage time was 318 (IQR 226 to 397) mins, indicating that CPAP was effective and adherence was good. CPAP adherence was negatively correlated with nasal resistance ($r = -0.188$, $p = 0.002$). The participants were divided into good ($n = 153$) and poor ($n = 107$) CPAP adherence groups. In the poor adherence group, rhinomanometry values before CPAP introduction were worse (inspiration, $p = 0.003$; expiration, $p = 0.006$). There was no significant difference in patient background when comparing those with improved ($n = 16$) and unimproved ($n = 12$) pre-CPAP high rhinomanometry values. However, CPAP usage time was significantly longer in the improved group 1 month ($p = 0.002$) and 3 months ($p = 0.026$) after CPAP introduction. The authors concluded that the findings of this study suggested that nasal resistance examined by rhinomanometry was a useful predictor of CPAP adherence, and that improved rhinomanometry values may contribute to extending the duration of CPAP use.

The authors stated that this study had several drawbacks. First, it was a retrospective study. Second, it was carried out in a single facility. Third, no nasal examination or anatomical findings were obtained, with nasal breathing disorders evaluated only by the nasal airflow test and total nasal symptom score. Fourth, manual titration was not carried out, and the setting pressure was changed depending on the patient's poor adherence status, which may have affected adherence. Fifth, these researchers evaluated nasal resistance by rhinomanometry in patients with OSA in the recumbent position, who were awake during the day without CPAP. To the authors' knowledge, no method has been established to measure nasal resistance during sleep or CPAP treatment, and determining a more reliable measurement method is essential. Therefore, in conjunction with the department of otorhinolaryngology, these researchers aimed to perform a prospective, multi-center study to examine the effect of various factors on improving CPAP adherence.

Maniaci et al (2024) noted that pediatric inferior turbinate hypertrophy (PedTH) is a frequent and often overlooked cause or associated cause of nasal breathing difficulties. This clinical consensus statement (CCS) aimed to provide a diagnosis and management frame-work covering the lack of specific guidelines for this condition and addressing the existing controversies. This CCS was developed by a panel of 20 contributors from 7 different European and North American countries using the modified Delphi method. The aim of the CCS was to offer a multi-disciplinary reference frame-work for the management of PedTH on the basis of shared clinical experience and analysis of the strongest evidence currently available. These investigators carried out a systematic literature review following the PRISMA guidelines. From the initial 96 items identified, 7 studies were selected based on higher-evidence items such as RCTs, guidelines, and systematic reviews. A 34-statement survey was developed, and after 3 rounds of voting, 2 items reached strong consensus, 17 reached consensus or near consensus, and 15 had no consensus. The authors concluded that until further prospective data are available, this CCS should provide a useful reference for PedTH management. PedTH should be considered a nasal obstructive disease not necessarily related to an adult condition but frequently associated with other nasal or cranio-facial disorders. Diagnosis requires clinical examination and endoscopy, whereas rhinomanometry, nasal cytology, and questionnaires have little clinical role. Level of Evidence = V.

Optical Rhinometry

Optical rhinometry (ORM) is a new technique introduced in Germany in 2004 that quantifies light extinction in optical density to assess nasal blood volume as a measure of nasal patency. It works via optical

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spectroscopy, which measures the absorption of visible and near-infrared light in tissue. Similar to pulse oximetry that measures hemoglobin absorption of near-infrared light and thus oxygen blood saturation, ORM measures blood volume within the nasal cavity. An emitter and detector are positioned across the nasal bridge, and swelling is measured as the extinction of light or optical density, as a function of time.

Hellgren and colleagues (2007) validated the Rhinolux (Rhios GmbH, Germany), an optical rhinometer, against AR in detecting nasal mucosal swelling when changing body position from sitting to supine. The study population consisted of 20 healthy subjects (7 women, 13 men, mean age of 34.7 +/- 9.3 years). The Rhinolux was applied sitting in the upright position followed by 5 mins in the supine position. Acoustic rhinometry was measured sitting in the upright position and after 5 mins in the supine position. In 7 subjects the measurements were repeated on 3 different days to assess the repeatability. The mean change from baseline in minimal cross sectional area DeltaMCA measured with acoustic rhinometry was -0.12 (+/- 0.19) cm2 (right + left side), p = 0.013 but DeltaE (change in light extinction from baseline) measured with the Rhinolux was unchanged 0.02 (+/- 0.18) optical densities (OD), p = 0.56. There was no correlation between DeltaE and DeltaMCA r = 0.028, p = 0.9. The mean DeltaE result from repeated measurements on different days was 0.05 (+/- 0.08) OD, p = 0.09 and the DeltaMCA was -0.1 (+/- 0.11) cm2, p = 0.02. This study showed that the changes in nasal blood volume measured with the Rhinolux did not reflect changes in nasal mucosal swelling measured with AR when changing body position from sitting to supine. The results indicated that the utility of the Rhinolux in assessing nasal mucosal reactions has to be evaluated further.

In a prospective pilot study, Cheung et al (2010) assessed ORM as an objective evaluation of nasal patency using NPT with Dermatophagoides farinae (Df) as compared with AR. A total of 5 adult healthy controls and 5 adult subjects with allergic rhinitis underwent NPT with increasing concentrations of Df while undergoing ORM. The minimum concentration of Df causing a positive reading was recorded. Nasal cross-sectional area was measured before and after testing using AR. Nasal patency was assessed subjectively after each challenge with the visual analog scale. The median amount of Df causing a positive response on ORM was less in allergic rhinitis patients as compared to healthy controls, at 5000 AU/ml and greater than 10,000 AU/ml, respectively. There was a statistically significant correlation between the change in optical density in ORM and subjective nasal congestion after increasing Df challenges (r = 0.63; p = 0.0007). Similarly, there was a statistically significant correlation between change in optical density by ORM and both minimum cross-sectional areas as measured by AR (r = -0.60, p = 0.03; and r = -0.64, p = 0.02, respectively). The authors concluded that this is the first study to show a correlation between ORM and AR during NPT with Df. In addition, the data support a correlation of ORM to subjective symptoms of nasal congestion. These findings suggest that ORM is able to assess changes in nasal patency during challenges with Df. They stated that further studies on ORM are needed; current ongoing trials are evaluating ORM for NPT with other common antigens.

Lambert et al (2013) noted that patients with non-allergic irritant rhinitis (NAIR) have symptoms of nasal congestion, nasal irritation, rhinorrhea, and sneezing in response to nasal irritants. There is currently no reliable objective means to quantify these patients' subjective symptoms. In this study, these researchers used the transient receptor potential vanilloid receptor (TRPV1) receptor agonist, capsaicin, as an intra-nasal challenge while comparing the changes in blood flow with optical rhinometry between subjects with NAIR and healthy controls (HCs). A total of 6 HCs and 6 NAIR subjects were challenged intra-nasally with saline solution followed by increasing concentrations of capsaicin (0.005 mM, 0.05 mM, and 0.5 mM) at 15-min intervals. These investigators recorded maximum optical density (OD) and numeric analog scores (NAS) for nasal congestion, nasal irritation, rhinorrhea, and sneezing for each subject after each challenge. Correlations between NAS and maximum OD were calculated. Maximum OD increased with increasing concentrations of intra-nasal capsaicin in NAIR subjects. There were significant differences in maximum OD obtained for 0.05 mM and 0.5 mM capsaicin between NAIR subjects and HCs. Significant differences were found in the NAS for nasal irritation at 0.005 mM, 0.05 mM, and 0.5 mM, and nasal congestion at 0.5 mM. Correlation between maximum OD and mean NAS was most significant for 0.05 mM capsaicin. The authors concluded that optical rhinometry with intra-nasal capsaicin challenge could prove a viable option in the diagnosis of NAIR. Moreover, they stated that further studies will investigate its use to monitor a patient's response to pharmacologic therapy and provide further information about the underlying mechanisms of NAIR. The findings of this small study need to be validated by well-designed studies.

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Krzych-Fałta and Samolinski (2016) noted that optical rhinometry is the only diagnostic tool in rhinitis for assessing real-time changes in nasal occlusion. It illustrates lumen changes of nasal mucosa vessels in response non-specific/specific factors and not only. The first attempts to standardize the method conducted by German researchers showed the potential of optical rhinometry not only as regards challenge tests, but also vice versa, in respect of the anemization of the mucosa it evaluates the extent of the edema that occurred in the patho-mechanism of non-allergic rhinitis. The authors concluded that the relatively small number of publications in the domain of interest demonstrated there is a need to conduct further research on the suitability of the above-mentioned technique for the evaluation of nasal patency in the field of rhinological diagnostics.

Patil and Jain (2024) noted that rhinomanometry is a diagnostic technique in rhinology, providing a quantitative assessment of nasal airflow and resistance. These investigators provided a comprehensive review that examined the historical development, principles, as well as clinical applications of rhinomanometry; emphasizing its role in diagnosing nasal obstructions, pre-operative evaluations and monitoring therapeutic outcomes. These researchers stated that despite facing challenges such as technical limitations and the need for standardization, rhinomanometry remains a tool in both clinical as well as research settings. In addition, the authors examined future directions, highlighting the potential for device miniaturization, telemedicine integration, personalized protocols and collaborative research efforts. These advancements will likely expand the accessibility, accuracy and clinical relevance of rhinomanometry, solidifying its role in the ongoing evolution of rhinology practice.

These researchers stated that current rhinomanometry devices have several technical limitations. The technical constraints entail various facets that affect their functionality and precision. A notable limitation lies in the difficulty of accurately correlating rhinomanometric data with the subjective perception of nasal patency, underscoring the necessity for enhanced calibration and measurement methodologies. Additionally, the use of active posterior rhinomanometry, while beneficial for evaluating nasal resistance, may encounter its own array of limitations, inaccuracies, and measurement challenges, especially when dealing with intricate nasal conditions such as rhinosinusitis, polyposis processes or septal deviations. Moreover, the disparities between in-vivo and in-silico measurements in rhinomanometry present a technical hurdle, with computational models often needing to catch up in predicting nasal resistance compared to real-world measurements, demonstrating a need for refined computational techniques and modelling approaches. The pragmatic constraints encountered during active anterior rhinomanometry, such as patient discomfort, potential artefacts stemming from mouth breathing as well as limitations during nasal provocation tests, further compounded the technical obstacles encountered by present-day rhinomanometry devices. In addition, current rhinomanometry devices also face interpretation challenges due to several factors. One notable challenge lies in the disparity between subjective symptoms of nasal obstruction and the objective measurements of nasal patency. While rhinomanometry provides insights into nasal resistance and air-flow, it may not consistently correlate with a patient's perceived degree of nasal blockage. Furthermore, the methodology of active anterior rhinomanometry, which involves evaluating nasal resistance during respiration, could encounter practical constraints during testing. These may include interruptions for allergen exposure, artefacts stemming from mouth breathing or issues related to mask fitting. Moreover, interpreting rhinomanometry findings necessitates skilled technicians and optimal patient cooperation, especially in posterior rhinomanometry. Ensuring controlled ambient conditions, achieving a snug seal of the facial mask and preventing mouth breathing are imperative for attaining reproducible and precise results. In addition, identifying abnormalities such as inflammation, nasal polyps or a deviated septum via rhinomanometry demands a careful evaluation by specialists to determine suitable treatments, which may encompass surgical or structural interventions. These researchers stated that patient-related factors could also significantly influence the outcomes of nasal air-flow assessments and measurements. Various physiological and pathological conditions could impact nasal air-flow and resistance, influencing diagnostic tests and evaluations. Variables such as the nasal cycle, septal deviations, turbinate hypertrophy, tumors, synechiae, nasal congestion, allergies, non-allergic rhinitis and sino-nasal polyposis could exert varying degrees of influence on air-flow quantity or nasal airway resistance; therefore, resulting in fluctuations in test outcomes. In addition, individual responses to decongestants offer insights into the underlying causes of nasal obstruction. A lack of response may signify structural obstructions such as septal deviations, bony hypertrophy of the turbinates, and inflammatory conditions unresponsive to decongestants. Furthermore, the capacity of patients to relax the soft palate during posterior rhinomanometry could significantly impact

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measurement accuracy, underscoring the criticality of patient cooperation and physiological factors in ensuring the reliability of results.

Im et al (2025) stated that OSA is a prevalent disorder associated with severe co-morbidities. Anatomical variations, including nasal diseases, have been identified as potential contributing factors. Ina retrospective study, these researchers examined the relationship between nasal CSAs and OSA, using both AR and CT. This trial was carried out on 355 individuals who underwent PSG, para-nasal sinus CT, and AR. Clinical characteristics were compared among the non-OSA, mild OSA, moderate OSA, and severe OSA patients. Multiple logistic regression was used to determine the influence of nasal CSAs on the risk of developing moderate or severe OSA, with adjustments for sex, age, and BMI. Significant associations between specific AR-measured CSA values of anterior nasal region and moderate-severe OSA were identified upon adjusting for confounders ($p = 0.034$, 0.010, and 0.025, respectively). The lower CSA measurements between the right and the left sides revealed a stronger correlation with moderate-severe OSA than the summed values, suggesting the importance of the narrower nasal passage in OSA severity. No significant impact of CSA values on the development of severe OSA was observed. The authors concluded that nasal CSA measured using AR appeared to be a better method for verifying its association with OSA than CT. These investigators stated that nasal CSAs at the anterior region of the inferior turbinates play a crucial role in the development of OSA, independent of other known risk factors. This was a retrospective study; its findings need to be validated by well-designed studies.

Agdogan and Ersozlu (2025) noted that rhinoplasty is a critical surgical intervention aimed at enhancing nasal form and function. However, traditional approaches often compromise the integrity of nasal scroll ligaments, vital for the functionality of the internal nasal valve, resulting in potential post-operative nasal dysfunction. Despite the importance of scroll ligaments in maintaining nasal structure and function, the literature lacks objective, quantifiable evidence of their role. In a retrospective, double-blind study, these investigators compared nasal functionality and anatomy between patients undergoing rhinoplasty with bilateral preservation of scroll ligaments and those with bilateral sacrifice of these ligaments, using computerized rhinomanometric and AR measurements as evaluative tools. This trial was carried out at Tekirdag Namık Kemal University's Plastic Reconstructive and Aesthetic Surgery and Ear, Nose, and Throat Clinics. The study population comprised patients who underwent rhinoplasty between May 1, 2018, and October 1, 2019. Participants were selected based on their medical history, including previous nasal surgeries, aesthetic dissatisfaction, as well as minimal nasal septum deviation. They were then categorized into 2 distinct groups: one undergoing rhinoplasty with an emphasis on preserving the scroll ligaments ($n = 31$) and the other where these ligaments were intentionally sacrificed ($n = 30$). These researchers conducted a comprehensive analysis involving pre-operative and 6-month post-operative computerized rhinomanometry and AR measurements to examine the surgical impact on nasal functionality and structure. The statistical comparison focused on assessing the changes in air pressure and air-flow rates, aiming to delineate the functional outcomes attributable to the differing surgical techniques. The results showed a stark contrast in post-operative nasal functionality between the 2 groups. In the group with preserved scroll ligaments, there was a significant improvement in post-operative computerized rhinomanometric measurements, indicating enhanced nasal air-flow and reduced airway resistance. In contrast, the group with sacrificed scroll ligaments showed deteriorated post-operative outcomes, with increased nasal airway resistance and decreased air-flow. These differences underscored the pivotal role of scroll ligaments in maintaining nasal airway patency and functionality. The authors concluded that the findings of this trial unequivocally showed that the preservation of scroll ligaments during rhinoplasty significantly benefited nasal airway function, as evidenced by improved post-operative rhinomanometry and AR measurements. These objective data provided a compelling argument for the functional importance of scroll ligaments, advocating for surgical techniques that prioritize their preservation. By highlighting the positive correlation between ligament preservation and enhanced nasal functionality, this study contributed insights into rhinoplasty practices, suggesting a paradigm shift towards approaches that safeguard nasal structural integrity for optimal functional outcomes. Level of Evidence = IV.

Castellanos et al (2025) noted that mouth breathing generates imbalances in the musculature, in cranio-facial morpho-functionality, and in the stomatognathic system; thus, it is important to diagnose mouth breathing via the quantitative assessment of nasal permeability, which can be carried out using rhinomanometry. In a systematic review, these investigators examined the effectiveness of rhinomanometry in the diagnosis of

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mouth breathing in pediatric patients. The guiding question was: Is the use of rhinomanometry as an assessment tool effective in the diagnosis of mouth breathing in pediatric patients?. These investigators carried out a search on the following data-bases: Latin American and Caribbean Center on Health Sciences Information (BIREME), Latin American and Caribbean Health Sciences Literature (LILACS), PubMed/Medical Literature Analysis and Retrieval System Online (MEDLINE), Scientific Electronic Library Online (SciELO), Web of Science, and Science Direct. The Health Sciences Descriptors and Medical Subjects Headings (MESH) were combined with the Boolean operator AND in the search strategy: rhinomanometry AND mouth breathing AND diagnosis AND nasal pressure AND nasal airflow AND nasal resistance . Observational cohort and cross-sectional studies that addressed the effectiveness of rhinomanometry in the diagnosis of mouth breathing were included. The reviewers independently extracted the information and scored the review quality based on the Physiotherapy Evidence Database (PEDro) scale and the grading of evidence levels according to the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system. Of the 1,536 studies identified, only 3 were selected for the present review following the application of the eligibility criteria. The authors concluded that the included studies in this systematic review employed rhinomanometry in their population of mouth breathers; however, there was a lack of standardization of this instrument to examine the efficiency of nasal resistance in the upper airways, as well as a methodological variability that decreased the reliability of the results found. Therefore, these investigators stated that it would be desirable to carry out more controlled studies in this field and with larger samples to obtain a better understanding of the consequences of mouth breathing on nasal permeability.

Maniam et al (2025) stated that the relationship between clinical examination findings and objective nasal patency measures in structural nasal obstruction remains uncertain. In a systematic review, these investigators examined the relationship between clinical nasal examination findings and objective nasal patency measures using AR, peak nasal inspiratory flow, rhinomanometry, as well as rhino-spirometry. They carried out a qualitative systematic review by means of the PRISMA 2020 statement. A total of 17 studies were included in the systematic review. Several studies showed a positive relationship between objective nasal patency measures and clinical nasal examination findings; however evidence in the literature is limited and confined to cohort studies. Objective nasal patency measures using AR, rhinomanometry and rhino-spirometry assessment correlated positively in severe anterior septal deviation; however, their roles in assessing middle/posterior and mild/moderate septal deviation in isolation remains uncertain. These investigators noted that most of the evidence described in this review was derived from low-level findings from cohort and case-series studies. The subjects from these cohort studies were heterogeneous and subjected to selection, publication and ascertainment bias. A meta-analysis was also not possible due to limited and heterogeneous evidence in the literature. The focus of the published literature has mainly been on septal deviation and other causes of nasal obstruction secondary to structural sinonasal disease such as nasal polyps and trauma remain unclear. The authors concluded that AR, rhinomanometry and rhino-spirometry assessment correlated positively in severe anterior septal deviation; however, their role in assessing middle/posterior and mild/moderate septal deviation in isolation remains uncertain. They stated that the evidence from this review highlighted that objective nasal patency measuring tools have a limited role in supporting clinical examination findings. Although objective nasal patency measures are useful in identifying patients with severe nasal obstruction, their use in examining mild or moderate nasal obstruction remains poor. Objective nasal patency measures can be used in conjunction with clinical examination findings to aid patient selection for septoplasty and predict post-operative satisfaction. Moreover, these researchers stated that further investigations are needed to examine how objective nasal patency measures can be used as an objective diagnostic tool when assessing severe, moderate and mild nasal obstruction.

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